

2. (Discrete Random Variables)

A variable X has the following probability mass function:

$$P(X=x) = \begin{cases} k(2x+1), & x=1, 2, 3, 4 \\ 0, & \text{otherwise} \end{cases}$$

a) Find the value of the constant k

$$k(2(1)+1) = k(3), \quad k(2(2)+1) = k(5) \quad k(2(3)+1) = k(7)$$

$$k(2(4)+1) = k(9)$$

$$\text{Probabilities: } P(1) = 3/24$$

$$P(2) = 5/24$$

$$P(3) = 7/24$$

$$P(4) = 9/24$$

$$k[(3) + (5) + (7) + (9)] = 1$$

$$\frac{24k}{24} = \frac{1}{24}$$

$$k = \frac{1}{24}$$

b) $E(X)$ and $\text{Var}(X)$

$$E(X) = 1\left(\frac{3}{24}\right) + 2\left(\frac{5}{24}\right) + 3\left(\frac{7}{24}\right) + 4\left(\frac{9}{24}\right)$$

$$= 2.9167 \quad \text{or} \quad \frac{35}{12}$$

$$E(X^2) = 1\left(\frac{3}{24}\right) + 4\left(\frac{5}{24}\right) + 9\left(\frac{7}{24}\right) + 16\left(\frac{9}{24}\right)$$

$$\text{Var}(X) = 9.5833 \quad \text{or} \quad \frac{115}{12}$$

$$\text{Var}(X) = E(X^2) - (E(X))^2 = \frac{115}{12} - \left(\frac{35}{12}\right)^2$$

$$= \frac{155}{144} \quad \text{or} \quad 1.0769$$

c) Calculate $P(X \geq 3)$

x $P(X=x)$

1 $3/24$

2 $5/24$

3 $7/24$

4 $9/24$

$$P(X \geq 3) = P(X=3) + P(X=4)$$

$$\frac{7}{24} + \frac{9}{24} = \frac{16}{24} \text{ or } \frac{2}{3}$$

$$P(X \geq 3) = \frac{2}{3} \text{ or } \boxed{0.6667}$$

2.) Binomial Distribution

An IT system detects that roughly 8% of incoming external requests are malicious. In a random sample of 20 requests

a) What is a probability that exactly 2 requests are malicious?

$$p = 0.08 \quad n = 20$$

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$\text{where } \binom{n}{x} = \frac{n!}{x!(n-x)!}$$

$$P(X=2) = \binom{20}{2} (0.08)^2 (0.92)^{18}$$

$$\binom{20}{2} = \frac{20!}{2!18!} = 190$$

$$(0.08)^2 = 0.0064$$

$$(0.92)^{18} = 0.222932$$

$$P(X=2) = 190 \times 0.0064 \times 0.222932$$

$$\approx \boxed{0.2711}$$

b) Probability of at least 3 requests are malicious

$$P(X \geq 3) = 1 - P(X < 3)$$

$$P(X=0) = \binom{20}{0} (0.08)^0 (0.92)^{20}$$

$$\binom{20}{0} = \frac{20!}{0!20!} = 1$$

$$(0.08)^0 = 1$$

$$(0.92)^{20} \approx 0.188695$$

$$P(X=0) = 1 \times 1 \times 0.188695 = 0.188695$$

$$P(X=1) = \binom{20}{1} (0.08)^1 (0.92)^{19}$$

$$\frac{20!}{1!19!} = 20$$

$$(0.08)^1 = 0.08$$

$$(0.92)^{19} \approx 0.205147$$

$$P(X=1) = 20 \times 0.08 \times 0.205147$$

$$= 0.328235$$

$$P(X=2) = 0.271089 \quad (\text{From a})$$

$$P(X \geq 3) = 1 - [0.188695 + 0.328235 + 0.271089]$$

$$P(X \geq 3) = 1 - 0.788019$$

$$= 0.211980$$

c.) Mean and SD of the number of malicious requests.

$$\mu = E[X] = np$$

$$\sigma = \sqrt{\text{Var}(X)} = \sqrt{np(1-p)}$$

where $n=20$, $p=0.08$, and $(1-p)=0.92$

$$\mu = np = 20 \times 0.08 = \boxed{1.6} \text{ MEAN}$$

$$\begin{aligned} \text{Var}(X) &= np(1-p) = 20 \times 0.08 \times 0.92 \\ &= 1.6 \times 0.92 = 1.472 \end{aligned}$$

$$\sigma = \sqrt{1.472} \quad \sigma \approx \boxed{1.2133} \text{ SD}$$

3) Poisson Distribution

A server receives an average of 4.5 web traffic hits per minute. Assuming the ~~that~~ arrival rate follows a Poisson process.

a) Receiving exactly 6 hits in a given minute.

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$\lambda = 4.5$ hits per minute

$$P(X=6) = \frac{e^{-4.5} (4.5)^6}{6!}$$

$$P(X=6) = \frac{(0.01110899695)(8303.765625)}{720}$$

$$e^{-4.5} = 0.01110899695$$

$$(4.5)^6 = 8303.765625$$

$$6! = 720$$

$$\approx \boxed{0.1281}$$

b) Receiving at most 2 hits in a given minute.

$$P(X \leq 2) = P(0) + P(1) + P(2)$$

$$P(X=0) = \frac{e^{-4.5} (4.5)^0}{0!} = \frac{0.011109 \times 1}{1} = 0.011109$$

$$P(X=1) = \frac{e^{-4.5} (4.5)^1}{1!} = \frac{0.011109 \times 4.5}{1} = 0.049990$$

$$P(X=2) = \frac{e^{-4.5} (4.5)^2}{2!} = \frac{0.011109 \times 20.25}{2} = 0.112478$$

$$P(X \leq 2) = 0.011109 + 0.049990 + 0.112478$$

$$= \boxed{0.1736}$$

c) Calculate probability of receiving more than 10 hits in a 2 min interval.

$$2 \text{ min interval: } \lambda = 4.5 \times 2 = 9$$

$$P(X > 10) = 1 - P(X \leq 10) = 1 - \sum_{x=0}^{10} P(X=x)$$

$$P(x) = P(x-1) \times \frac{\lambda}{x}$$

$$x=0$$

$$P(0) = \frac{e^{-9} (9)^0}{0!} = \frac{0.00012341 \times 1}{1} = 0.00012341$$

$$x=1$$

$$P(1) = \frac{e^{-9} (9)^1}{1!} = \frac{0.00012341 \times 9}{1} = 0.00111069$$

$$x = 2$$

$$P(2) = \frac{0.00012391 \times 81}{2} = \frac{0.0099621}{2} = 0.0049811$$

$$2^2 = 81$$

$$2! = 2$$

$$x = 3 \quad P(3) = \frac{0.00012391 \times 729}{6} = 0.0199932$$

$$3^3 = 729$$

$$3! = 6$$

$$x = 4 \quad P(4) = \frac{0.00012391 \times 6561}{24} = \frac{0.80969301}{24} = 0.03373721$$

$$4^4 = 6561$$

$$4! = 24$$

$$x = 5 \quad P(5) = \frac{0.00012391 \times 59049}{120} = 0.06072698$$

$$5^5 = 59049$$

$$5! = 120$$

$$x = 6 \quad P(6) = \frac{0.00012391 \times 531441}{720} = 0.09109047$$

$$6^6 = 531441$$

$$6! = 720$$

$$x = 7 \quad P(7) = \frac{0.00012391 \times 4782969}{5040} = 0.11711234$$

$$7^7 = 4782969$$

$$7! = 5040$$

$$x = 8 \quad P(8) = \frac{0.00012391 \times 43046721}{40320} = 0.13174593$$

$$8^8 = 43046721$$

$$8! = 40320$$

$$x = 9 \quad P(9) = \frac{0.00012391 \times 387420489}{362880} = 0.13174593$$

$$9^9 = 387420489$$

$$9! = 362880$$

$$x = 10 \quad P(10) = \frac{0.00012391 \times 3486784401}{3628800} = 0.11857134$$

$$10^{10} = 3486784401$$

$$10! = 3628800$$

$$P(X \leq 10) = 0.00012391 + 0.0011069 + 0.00499811 + 0.01999932 + 0.03373721$$

$$+ 0.06072698$$

$$\text{TOTAL: } P(X \leq 10) \approx 0.70596$$

$$1 - 0.70596 = P(X > 10) = 0.2940$$

4.) Continuous Probability & Normal Distribution

The processing time for a complex database query is normally distributed with a mean $\mu = 120$ milliseconds and a SD $\sigma = 15$ milliseconds

a.) Proportion of queries take longer than 135 milliseconds to process

$$Z = \frac{X - \mu}{\sigma} \quad P(X > 135)$$

$$Z = \frac{135 - 120}{15} = \frac{15}{15} = 1.00 \quad Z \text{ value: } 0.8913$$

$$P(X > 135) = 1 - 0.8913 = 0.1587$$

15.87% of database queries take longer than 135 milliseconds.

b.) Probability a query takes between 100 and 130 ms

Lower bound ($X=100$)

$$Z = \frac{100 - 120}{15} = \frac{-20}{15} = -1.3333$$

Z value: 0.09176

Upper bound ($X=130$)

$$Z = \frac{130 - 120}{15} = \frac{10}{15} = 0.6667$$

Z value: 0.74857

$$= 0.74857 - 0.09176 = 0.65681 \approx 65.68\%$$

About 65.68% of database queries finish between 100 and 130 ms.

c.) Find the 95th percentile of query processing times.

$$P(X \leq x_{0.95}) = 0.95$$

Z score that gives closest to 0.9500:

$$1.64 \approx 0.9495$$

$$1.65 \approx 0.9505$$

$$\frac{1.64 + 1.65}{2} = 1.645$$

$$Z = \frac{X - \mu}{\sigma} \quad X = \mu + Z\sigma$$

$$\mu = 120, \sigma = 15, Z = 1.645$$

$$\begin{aligned} X_{0.95} &= 120 + (1.645)(15) \\ &= 120 + 24.675 \\ &= \boxed{144.675 \text{ ms}} \end{aligned}$$

95% of database finish in 144.675 ms or less.

